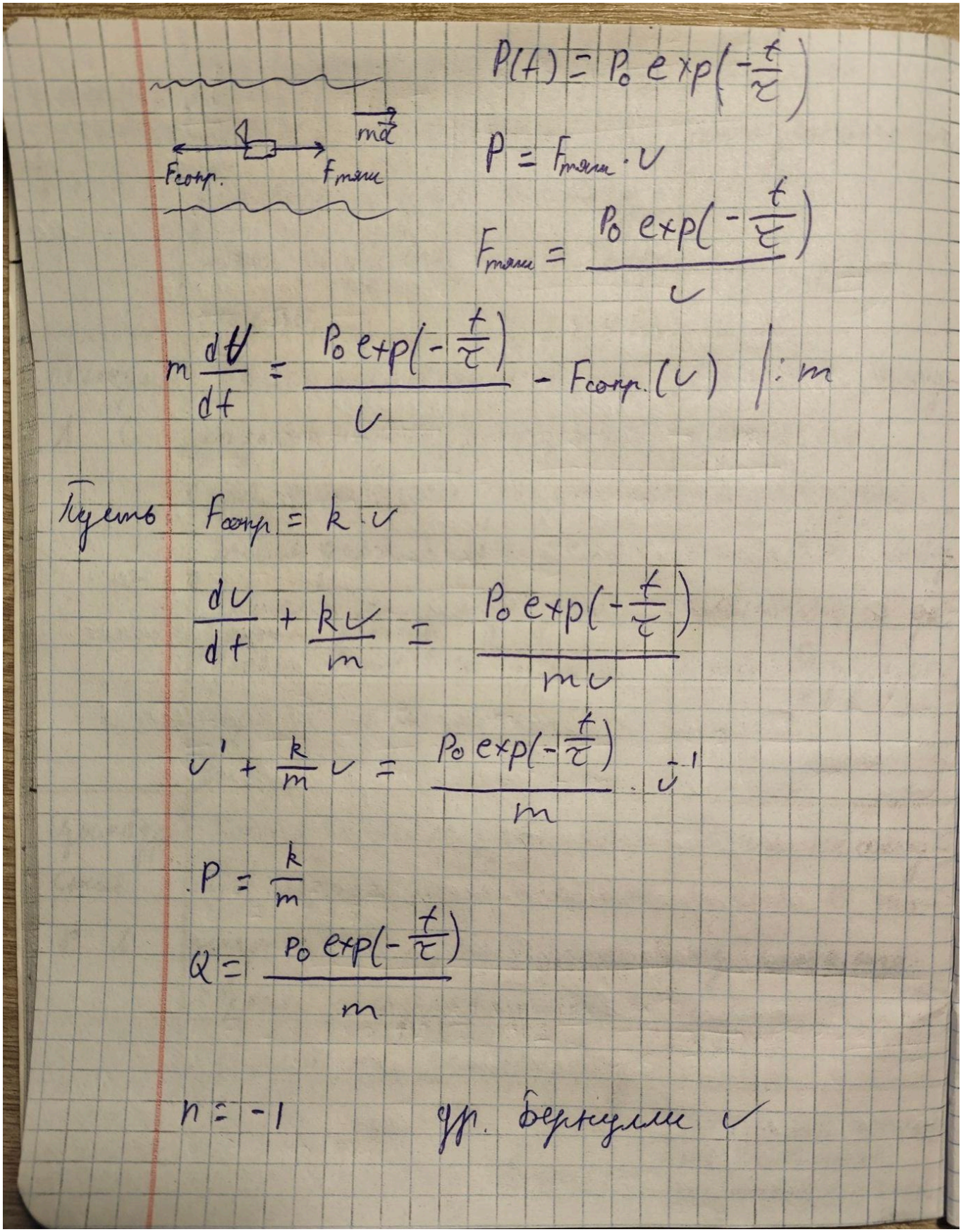
Задача про мотор



$$P(t) = P_0 \exp\left(-\frac{t}{\tau}\right)$$

$$P = F_{тяги} \cdot v$$

$$F_{тяги} = \frac{P_0 \exp\left(-\frac{t}{\tau}\right)}{v}$$

$$m\frac{dv}{dt} = \frac{P_0 \exp\left(-\frac{t}{\tau}\right)}{v} - F_{сопр.}(v) \quad | : m$$

Пусть $F_{сопр.} = k \cdot v$

$$\frac{dv}{dt} + \frac{kv}{m} = \frac{P_0 \exp\left(-\frac{t}{\tau}\right)}{mv}$$

$$v' + \frac{k}{m}v = \frac{P_0 \exp\left(-\frac{t}{\tau}\right)}{m} \cdot v^{-1}$$

$$P = \frac{k}{m}$$

$$Q = \frac{P_0 \exp\left(-\frac{t}{\tau}\right)}{m}$$

$n = -1$ ур. Бернулли ✓

$y = uv \quad \rightarrow \quad v = ls$

$y' = u'v + uv' \quad \rightarrow \quad v' = l's + ls'$

$$l's + ls' + \frac{k}{m} ls = \frac{P_0 \exp(-\frac{t}{\tau})}{m} \cdot \frac{1}{ls}$$

$$l's + l(s' + \frac{k}{m} s) = \frac{P_0 \exp(-\frac{t}{\tau})}{m} \cdot \frac{1}{ls}$$

$$s' + \frac{k}{m} s = 0 \quad | : s$$

$$\int \frac{ds}{s} = -\int \frac{k}{m} dt$$

$$\ln s = -\frac{k}{m} t$$

$$s = \exp(-\frac{k}{m} t) = e^{-\frac{kt}{m}}$$

$$l' \exp(-\frac{kt}{m}) = \frac{P_0 \exp(-\frac{t}{\tau})}{m} \cdot \frac{\exp(\frac{kt}{m})}{l}$$

$$l' = \frac{P_0 \exp(\frac{2kt}{m} - \frac{t}{\tau})}{ml}$$

$$\int l\,dl = \int \frac{P_0 \exp\left(\frac{2kt}{m} - \frac{t}{\tau}\right)}{m} dt$$

$$\frac{l^2}{2} = \frac{P_0}{m} \int e^x \cdot \frac{m\tau}{2k\tau - m} dx \quad (=)$$

$$x = \frac{2kt - \frac{tm}{\tau}}{m} = \frac{2k\tau t - tm}{m\tau}$$

$$dx = \frac{2k\tau - m}{m\tau} dt$$

$$(=)\ \frac{l^2}{2} = \frac{P_0}{m} \cdot \frac{m\tau}{2k\tau - m} \cdot \exp\left(\frac{2k\tau t - tm}{m\tau}\right) + C$$

$$v^2 = l^2 \cdot s^2$$

$$v = \pm \sqrt{\exp\left(-2\frac{kt}{m}\right) \cdot 2\frac{P_0}{m} \cdot \frac{m\tau}{2k\tau - m} \cdot \exp\left(\frac{2k\tau t - t\frac{m}{m\tau}}{m\tau}\right) + C}$$

$v > 0$ по усл.

$$v = \sqrt{\frac{2P_0\tau}{2k\tau - m} \cdot \exp\left(\frac{t(2k\tau - m)}{m\tau} - 2\frac{kt}{m}\right) + C}$$

$v(0) = v_0$

$$v_0 = \sqrt{\frac{2P_0\tau}{2k\tau - m} + C}$$

$$v_0^2 = \frac{2P_0\tau}{2k\tau - m} + C \rightarrow C = v_0^2 - \frac{2P_0\tau}{2k\tau - m}$$

$$v = \sqrt{\frac{2P_0\tau}{2k\tau - m}\left(\exp\left(\frac{t(2k\tau - m)}{m\tau} - 2\frac{kt}{m}\right) - 1\right) + v_0^2}$$

$\left.\frac{n}{\cdot}\right) + C$

$\frac{t-t}{m\tau}\frac{m}{}\Big) + C$

Задача про популяцию

$$\frac{N'}{N} = \frac{1}{N}\frac{dN}{dt}$$

прирост на 1 ед.

$$r(t) = a - bt \quad ; a, b > 0$$

$$\frac{dN}{dt} = -cN^2, \quad c > 0$$

$$\frac{dN}{dt} = (a-bt)N - cN^2$$

замедление

изменение популяции

прирост от кол-во ед.

$$N' - (a-bt)N = -cN^2$$

$P = -(a-bt)$ ур. Бернулли

$Q = -c$

$n = 2$

$N = uv$

$N' = u'v + uv'$

$$u'v + uv' - (a-bt)uv = -cu^2v^2$$

$$u'v + u(v' - (a - bt)v) = -c\,u^2 v^2$$

$$v' - (a - bt)v = 0 \quad | : v$$

$$\int \frac{dv}{v} = \int a - bt\,dt$$

$$\ln v = at - \frac{bt^2}{2}$$

$$v = e^{at - \frac{bt^2}{2}}$$

$$u' e^{at - \frac{bt^2}{2}} = -c\,u^2 \cdot e^{2at - bt^2} \quad | : u^2$$

$$\int \frac{du}{u^2} = \int -c \cdot e^{2at - bt^2 - at + \frac{bt^2}{2}}\,dt$$

$$-\frac{1}{u} = -c \int e^{at - \frac{bt^2}{2}}\,dt$$

$$u = \frac{1}{c \int \exp\left(at - \frac{bt^2}{2}\right) dt}$$

$$x = at - \frac{bt^2}{2}$$

$$dx = a - bt\ dt$$

$$\mu = \frac{e^{at - \frac{bt^2}{2}}}{c \int \exp\left(at - \frac{bt^2}{2}\right) dt}$$